Lecture 4: Policy and Policy Causality

Christian Wolf

MIT

14.461, Fall 2026

This lecture

- Objective: better understand role of **policy** in **impulse-propagation paradigm**
- Will proceed in three steps:
  1. **History of thought**: Cowles commission, Lucas critique, Sims program, ...
     Standard view: clear distinction between policy "shocks" and systematic policy rule design.
  2. New **shocks-rules equivalence**. Reduce role of policy to two margins:
     a) On-equilibrium causal effects of the policy instrument
     b) Equilibrium selection
  3. Towards empirics: **policy causal effects** in theory & data
- Overall: substantiate Lecture 1's intuitive discussion of **policy causality**

Outline

1. History of Thought
   - Shocks vs. Rules in the Literature
   - Towards a Bridge: Sims-Zha

2. Shocks-Rules Equivalence
   - Identification Results
   - Connection to Lucas & Sims

3. Policy Causal Effects in Theory & Data

Outline

1. History of Thought
   - Shocks vs. Rules in the Literature
   - Towards a Bridge: Sims-Zha

2. Shocks-Rules Equivalence
   - Identification Results
   - Connection to Lucas & Sims

3. Policy Causal Effects in Theory & Data

Policy evaluation in macroeconomics

- Dominant methodological tradition post-1950s: **Cowles Commission approach**
  - Basic idea: model economy as system of interdependent structural equations, with coefficients estimated using ML/IV methods
    E.g., postulate aggregate consumption function, estimate using lagged macro var's as IVs.
  - Close the model with policy. For counterfactuals, change policy & re-solve.
- Fundamental change: **rational-expectations revolution**/**Lucas critique**

  "Given that the structure of an econometric model consists of optimal decision rules of economic agents, and that optimal decision rules vary systematically with changes in the structure of series relevant to the decision maker, it follows that any change in policy will systematically alter the structure of econometric models."—Lucas (1976)
- Leads to a bifurcation in methodology and focus ...

Outline

1. History of Thought
   - Shocks vs. Rules in the Literature
   - Towards a Bridge: Sims-Zha

2. Shocks-Rules Equivalence
   - Identification Results
   - Connection to Lucas & Sims

3. Policy Causal Effects in Theory & Data

Policy shocks

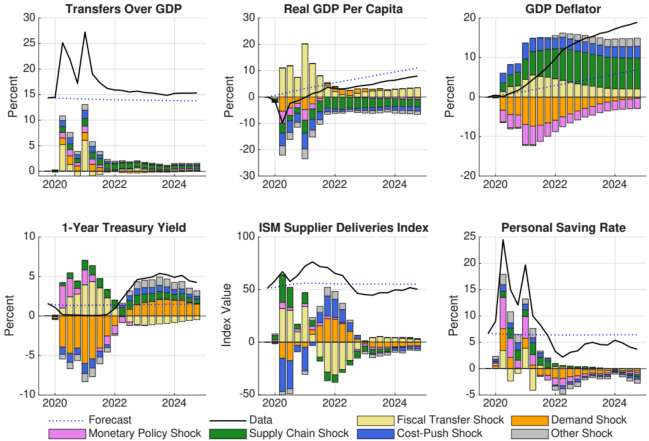
- Tradition started by Sims (1980): study **policy shocks**
  - Remember the monetary policy rule in the textbook NK model:

$$i_t = \phi_\pi \pi_t + \nu_t \tag{1}$$

    Note: slight notation switch, will be useful later.

  - Suppose you can somehow identify ν_t. Then you can:
    a) Estimate the causal effects of ν_t on the macro-economy
       Remember from Lecture 1: those are just the Θ_ℓ's for the monetary policy shock ν_t.
    b) Evaluate the contribution of ν_t to any given historical episode
  - Scope is deliberately limited: this is about surprise shocks to policy, *not* changes in the rule
    Rule changes revise private-sector decision rules, shocks do not.
- Next slide: typical shock exercise, gives contribution of ν_t to post-2021 inflation

The contribution of monetary shocks post-2021



Source: Arias, Rubio-Ramírez, and Shin (2026)

Policy rules

- Lucas program: counterfactual policy evaluation through **structural models**
  - Specify a complete structural model, solved as in Lecture Note 2, with shocks e_t:

$$\begin{aligned} s_t &= As_{t-1} + Be_t \\ y_t &= Cs_{t-1} + De_t \end{aligned}$$

    Policy shocks in e_t. Changing the policy *rule* changes all of $\{A, B, C, D\}$.
  - Strategy: discipline model to match historical episode of interest, then change policy [and thus A, B, C, D], finally re-solve
- Limited role of **policy shocks**: model validation

> *"[Economists] need to test them [models] as useful imitations of reality by subjecting them to shocks for which we are fairly certain how actual economies or parts of economies would react. The more dimensions on which the model mimics the answers actual economies give to simple questions, the more we trust its answers to harder questions."—Lucas (1980)*

Policy counterfactual questions

- Given a model, we can evaluate several interesting **policy counterfactuals**

  Q1 **Shock propagation**: how is the propagation of a particular shock affected?
  E.g., how would oil shocks propagate in the absence of a central bank reaction?

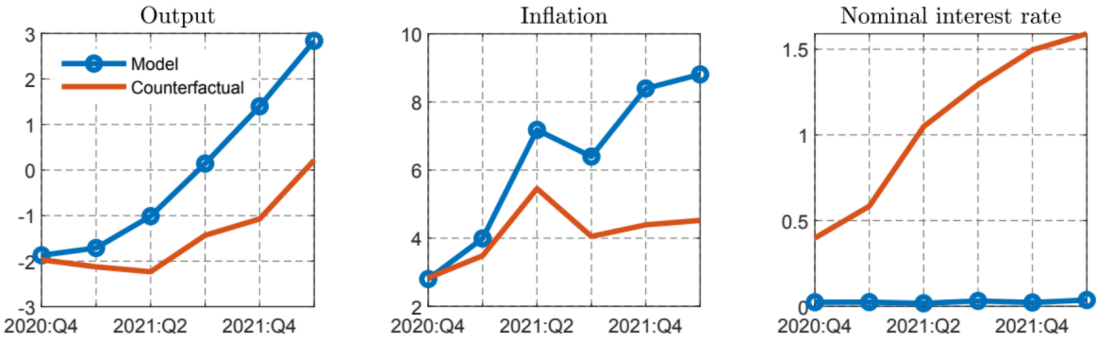
  Q2 **Business-cycle statistics**: how different would the average business cycle look?
  E.g., how would strict adherence to a Taylor rule have shaped output volatility post-WWII?

  Q3 **Alternative history**: how would a particular historical episode have unfolded?
  E.g., how would the post-covid inflation have unfolded with earlier rate hikes?

  Note: all of those are viewed as fundamentally distinct from the propagation of **policy shocks**.

- Next slide: typical systematic policy counterfactual exercise, changing **interest rates** during the post-2021 inflation

Changing interest rates post-2021



Source: Bocola, Dovis, Jørgensen, and Kirpalani (2024)

Outline

1. History of Thought
   - Shocks vs. Rules in the Literature
   - Towards a Bridge: Sims-Zha

2. Shocks-Rules Equivalence
   - Identification Results
   - Connection to Lucas & Sims

3. Policy Causal Effects in Theory & Data

Can we go from shock effects to rule changes?

- Sims and Zha (1998, 2006) proposed a strategy to use **policy shocks** to evaluate changes in **systematic policy design**
- Not particularly well-received – here for **Q1**:

> *"Sims and Zha [...] argue that if the analyst has made enough assumptions to isolate another fundamental shock to the economy, then it is possible to understand the consequences of a change in the systematic way that monetary policy responds to that shock, even in the absence of a structural model. Their arguments depend in a critical way on ignoring the Lucas critique. [...] We are open minded but skeptical."—Christiano, Eichenbaum, & Evans (1999)*

- Our starting point: try to replicate & understand their proposed strategy
  1. Present a (simplified version of) the Sims-Zha approach
  2. Using models as laboratories, try to understand when this approach may work

From there we can build towards sharp identification results that connect shocks & rules ...

Sims-Zha

Illustration using Q1: **shock propagation** w/ counterfactual monetary policy

Concretely: how would shock ε_t have propagated if the rule had been $i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$?

Sims-Zha

Illustration using Q1: **shock propagation** w/ counterfactual monetary policy

Concretely: how would shock $\boldsymbol{\varepsilon_t}$ have propagated if the rule had been $i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$?

- Sims-Zha is an empirical, model-free way to give an answer to Q1. Required **inputs**:
  1. IRFs to the (private-sector/non-policy) shock $\boldsymbol{\varepsilon_t}$ under the baseline, prevailing policy rule

     $$\{\boldsymbol{i}^{\varepsilon}, \boldsymbol{\pi}^{\varepsilon}, \boldsymbol{y}^{\varepsilon}\}$$

     Each of these is a $T \times 1$ IRF vector. Can estimate using regressions on an observed shock $\boldsymbol{\varepsilon_t}$.

  2. IRFs to a monetary policy shock $\boldsymbol{\nu_t}$

     $$\{\boldsymbol{i}^{\nu}, \boldsymbol{\pi}^{\nu}, \boldsymbol{y}^{\nu}\}$$

     Each of these is a $T \times 1$ IRF vector. Can estimate using regression on a single monetary shock $\boldsymbol{\nu_t}$.

Sims-Zha

Illustration using **Q1**: **shock propagation** w/ counterfactual monetary policy

Concretely: how would shock $\boldsymbol{\varepsilon_t}$ have propagated if the rule had been $\boldsymbol{i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t}$?

- Note that these **inputs** can be recovered purely from data. How do they then get used?

1. Choose a policy shock ν_0 at date 0 to enforce the **counterfactual rule**:

$$i_0^\varepsilon + \nu_0 \times i_0^\nu = \tilde{\phi}_\pi \times [\pi_0^\varepsilon + \nu_0 \times \pi_0^\nu] + \tilde{\phi}_y \times [y_0^\varepsilon + \nu_0 \times y_0^\nu]$$

This is one equation in one unknown.

Sims-Zha

Illustration using **Q1**: **shock propagation** w/ counterfactual monetary policy

Concretely: how would shock $\boldsymbol{\varepsilon_t}$ have propagated if the rule had been $i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$?

- Note that these **inputs** can be recovered purely from data. How do they then get used?

1. Choose a policy shock ν_0 at date 0 to enforce the **counterfactual rule**:

$$i_0^\varepsilon + \nu_0 \times i_0^\nu = \tilde{\phi}_\pi \times [\pi_0^\varepsilon + \nu_0 \times \pi_0^\nu] + \tilde{\phi}_y \times [y_0^\varepsilon + \nu_0 \times y_0^\nu]$$

   This is one equation in one unknown.

2. Iteratively continue for all $t = 1, 2, \ldots$. For $t = 1$:

$$i_1^\varepsilon + \nu_0 \times i_1^\nu + \nu_1 \times i_0^\nu = \tilde{\phi}_\pi \times [\pi_1^\varepsilon + \nu_0 \times \pi_1^\nu + \nu_1 \times \pi_0^\nu] + \tilde{\phi}_y \times [y_1^\varepsilon + \nu_0 \times y_1^\nu + \nu_1 \times y_0^\nu]$$

Sims-Zha

Illustration using **Q1**: **shock propagation** w/ counterfactual monetary policy

Concretely: how would shock $\boldsymbol{\varepsilon_t}$ have propagated if the rule had been $i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$?

- Note that these **inputs** can be recovered purely from data. How do they then get used?

1. Choose a policy shock ν_0 at date 0 to enforce the **counterfactual rule**:

$$i_0^\varepsilon + \nu_0 \times i_0^\nu = \tilde{\phi}_\pi \times [\pi_0^\varepsilon + \nu_0 \times \pi_0^\nu] + \tilde{\phi}_y \times [y_0^\varepsilon + \nu_0 \times y_0^\nu]$$

   This is one equation in one unknown.

2. Iteratively continue for all $t = 1, 2, \ldots$. For $t = 1$:

$$i_1^\varepsilon + \nu_0 \times i_1^\nu + \nu_1 \times i_0^\nu = \tilde{\phi}_\pi \times [\pi_1^\varepsilon + \nu_0 \times \pi_1^\nu + \nu_1 \times \pi_0^\nu] + \tilde{\phi}_y \times [y_1^\varepsilon + \nu_0 \times y_1^\nu + \nu_1 \times y_0^\nu]$$

3. Compute IRFs of everything to $\boldsymbol{\varepsilon}$ plus the *sequence* of policy shocks $\{\boldsymbol{\nu_0, \nu_1, \nu_2, \ldots}\}$.

Sims-Zha

Illustration using Q1: **shock propagation** w/ counterfactual monetary policy

Concretely: how would shock $\boldsymbol{\varepsilon_t}$ have propagated if the rule had been $i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$?

- Note that these **inputs** can be recovered purely from data. How do they then get used?

1. Choose a policy shock ν_0 at date 0 to enforce the **counterfactual rule**:

$$i_0^\varepsilon + \nu_0 \times i_0^\nu = \tilde{\phi}_\pi \times [\pi_0^\varepsilon + \nu_0 \times \pi_0^\nu] + \tilde{\phi}_y \times [y_0^\varepsilon + \nu_0 \times y_0^\nu]$$

   This is one equation in one unknown.

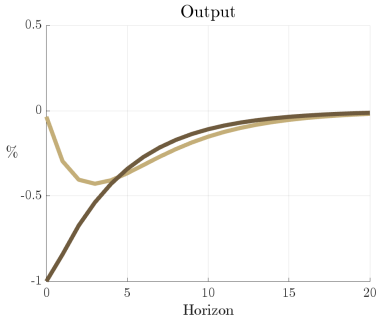
2. Iteratively continue for all $t = 1, 2, \ldots$. For $t = 1$:

$$i_1^\varepsilon + \nu_0 \times i_1^\nu + \nu_1 \times i_0^\nu = \tilde{\phi}_\pi \times [\pi_1^\varepsilon + \nu_0 \times \pi_1^\nu + \nu_1 \times \pi_0^\nu] + \tilde{\phi}_y \times [y_1^\varepsilon + \nu_0 \times y_1^\nu + \nu_1 \times y_0^\nu]$$

3. Compute IRFs of everything to $\boldsymbol{\varepsilon}$ plus the *sequence* of policy shocks $\{\boldsymbol{\nu_0, \nu_1, \nu_2, \ldots}\}$.

Next slide: illustration of what this method recovers when the true DGP is a standard macro model.

Interpreting the Sims-Zha estimand

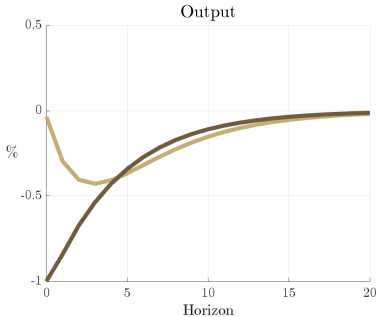


Experiment: cost-push shock under **baseline rule** & **cnfct'l rule** $i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$

These are the true IRFs in some linearized HANK model. For now the details don't matter.

Interpreting the Sims-Zha estimand

Q: What would an econometrician living in this model recover if she followed Sims-Zha?

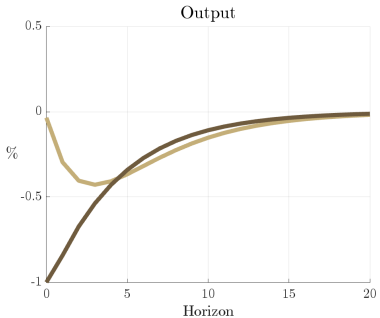


Experiment: cost-push shock under **baseline rule** & **cnfct'l rule** $i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$

These are the true IRFs in some linearized HANK model. For now the details don't matter.

Interpreting the Sims-Zha estimand

Q: What would an econometrician living in this model recover if she followed Sims-Zha?

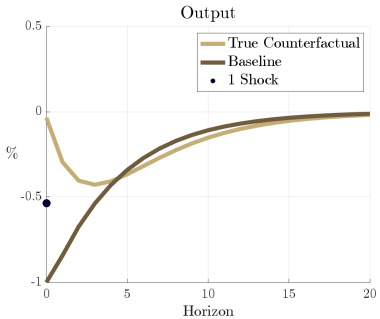


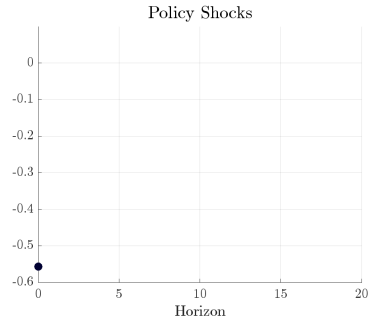
- Sims-Zha: use **ε IRFs under baseline policy rule** + **policy shock IRFs**

Strategy: enforce **counterfactual rule** using sequence of one-time **monetary shocks**

Interpreting the Sims-Zha estimand

Q: What would an econometrician living in this model recover if she followed Sims-Zha?





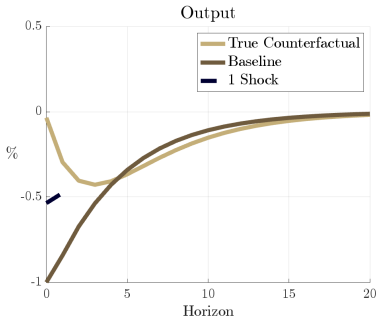
- Sims-Zha: use $\boldsymbol{\varepsilon}$ **IRFs under baseline policy rule** + **policy shock IRFs**
- Set MP shocks to values $\{\nu_0, \nu_1, \nu_2, \dots\}$ so that $\{i_t, \pi_t, y_t\}$ are related as

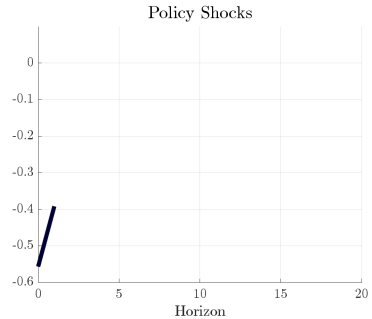
$$i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$$

Strategy: enforce **counterfactual rule** using sequence of one-time **monetary shocks**

Interpreting the Sims-Zha estimand

Q: What would an econometrician living in this model recover if she followed Sims-Zha?





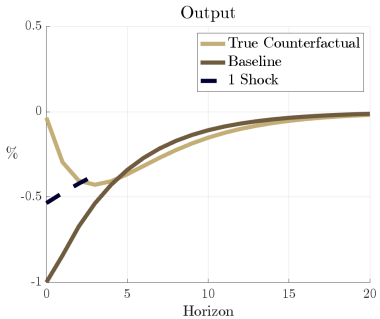
- Sims-Zha: use $\boldsymbol{\varepsilon}$ **IRFs under baseline policy rule** + **policy shock IRFs**
- Set MP shocks to values $\{\nu_0, \nu_1, \nu_2, \dots\}$ so that $\{i_t, \pi_t, y_t\}$ are related as

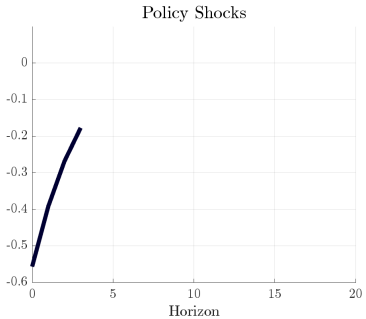
$$i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$$

Strategy: enforce **counterfactual rule** using sequence of one-time **monetary shocks**

Interpreting the Sims-Zha estimand

Q: What would an econometrician living in this model recover if she followed Sims-Zha?





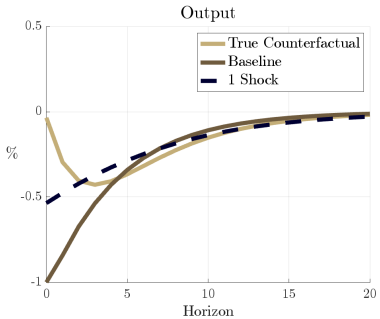
- Sims-Zha: use $\boldsymbol{\varepsilon}$ **IRFs under baseline policy rule** + **policy shock IRFs**
- Set MP shocks to values $\{\nu_0, \nu_1, \nu_2, \dots\}$ so that $\{i_t, \pi_t, y_t\}$ are related as

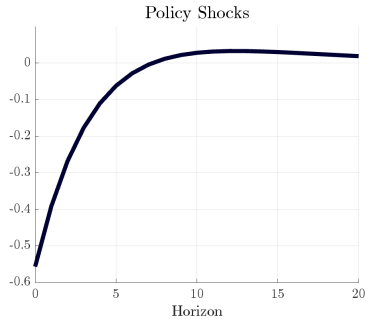
$$i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$$

Strategy: enforce **counterfactual rule** using sequence of one-time **monetary shocks**

Interpreting the Sims-Zha estimand

Q: What would an econometrician living in this model recover if she followed Sims-Zha?





- Sims-Zha: use $\boldsymbol{\varepsilon}$ **IRFs under baseline policy rule** + **policy shock IRFs**
- Set MP shocks to values $\{\nu_0, \nu_1, \nu_2, \dots\}$ so that $\{i_t, \pi_t, y_t\}$ are related as

$$i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$$

Strategy: enforce **counterfactual rule** using sequence of one-time **monetary shocks**

Problem: at each t, private sector expects return to old rule from $t+1$ onwards, so $\neq$ true cnfct'l.

Outline

1. History of Thought
   - Shocks vs. Rules in the Literature
   - Towards a Bridge: Sims-Zha

2. Shocks-Rules Equivalence
   - Identification Results
   - Connection to Lucas & Sims

3. Policy Causal Effects in Theory & Data

Building on Sims-Zha

Concern so far: using one-off policy shocks misses **expectational effects**

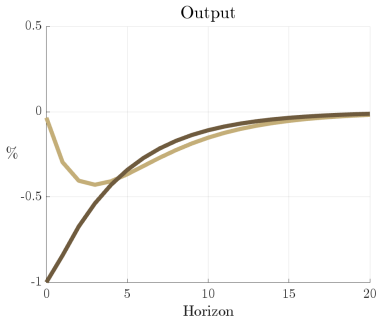
- Solution idea: use *multiple distinct* **policy shocks**
  - Concrete example: consider the monetary policy rule + shocks

$$i_t = f(\Omega_t) + \nu_{0,t} + \sum_{\ell=1}^{\infty} \nu_{\ell,t-\ell}$$

    where Ω_t is the date-t information set, $f(\bullet)$ is the systematic policy rule, and we suppose our econometrician is able to estimate the effects of the first n policy shocks
    Notation: $\nu_{\ell,t}$ is the date-t shock that changes policy at date $t+\ell$.
  - Now we have more degrees of freedom: we could enforce the counterfactual rule ex post in eq'm, but also in ex ante expectation for $n-1$ periods
- **Q**: does that get us closer to the truth? what happens as n gets large?

More graphical explorations

Q: What would the econometrician recover if she followed our generalization of Sims-Zha?



- Info: get IRFs to MP shocks

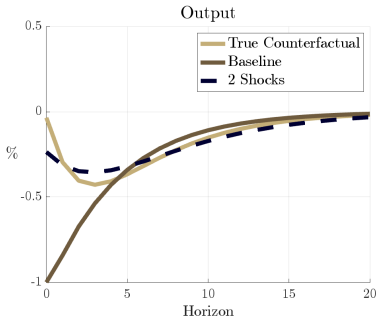
  $i_t = f(\Omega_t) + \nu_{0,t} + \nu_{1,t-1} + \dots$

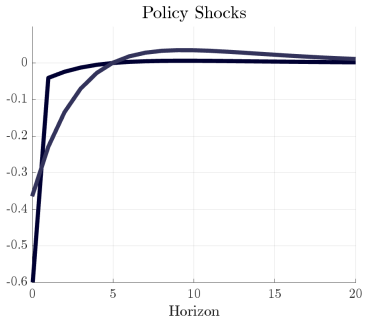
Insight: with **multiple monetary shocks** we can start to also match *expectations*

With n shocks we can enforce the rule today and in expectation for the next $n-1$ periods.

More graphical explorations

Q: What would the econometrician recover if she followed our generalization of Sims-Zha?





- Info: get IRFs to MP shocks

  $i_t = f(\Omega_t) + \nu_{0,t} + \nu_{1,t-1} + \dots$

- Set MP shocks so that $\{i_t, \pi_t, y_t\}$ are related as

  $$i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$$

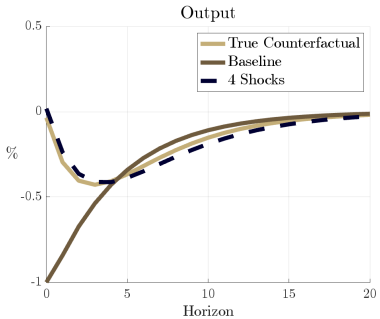
  not just ex post on eq'm path but also *in expectation*

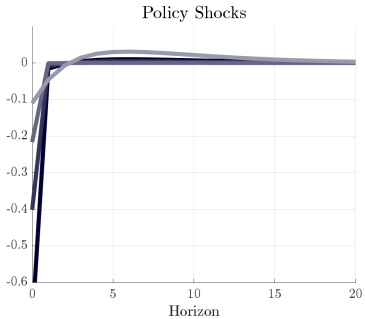
Insight: with **multiple monetary shocks** we can start to also match *expectations*

With n shocks we can enforce the rule today and in expectation for the next $n-1$ periods.

More graphical explorations

Q: What would the econometrician recover if she followed our generalization of Sims-Zha?





- Info: get IRFs to MP shocks

  $i_t = f(\Omega_t) + \nu_{0,t} + \nu_{1,t-1} + \dots$

- Set MP shocks so that $\{i_t, \pi_t, y_t\}$ are related as

  $$i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$$

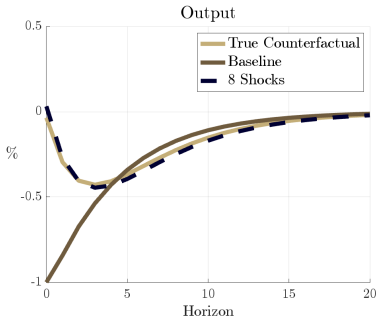
  not just ex post on eq'm path but also *in expectation*

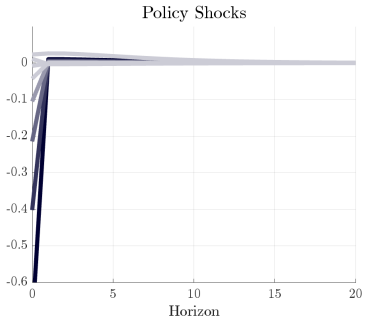
Insight: with **multiple monetary shocks** we can start to also match *expectations*

With n shocks we can enforce the rule today and in expectation for the next $n-1$ periods.

More graphical explorations

Q: What would the econometrician recover if she followed our generalization of Sims-Zha?





- Info: get IRFs to MP shocks

  $$i_t = f(\Omega_t) + \nu_{0,t} + \nu_{1,t-1} + \dots$$

- Set MP shocks so that $\{i_t, \pi_t, y_t\}$ are related as

  $$i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$$

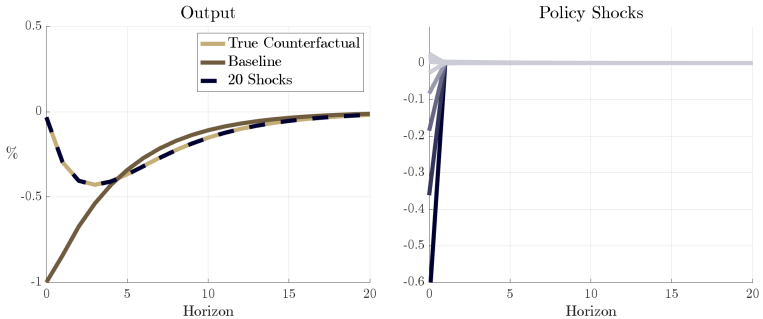
  not just ex post on eq'm path but also *in expectation*

Insight: with **multiple monetary shocks** we can start to also match *expectations*

With n shocks we can enforce the rule today and in expectation for the next $n-1$ periods.

More graphical explorations

Q: What would the econometrician recover if she followed our generalization of Sims-Zha?



- Info: get IRFs to MP shocks

  $i_t = f(\Omega_t) + \nu_{0,t} + \nu_{1,t-1} + \dots$

- Set MP shocks so that $\{i_t, \pi_t, y_t\}$ are related as

  $$i_t = \tilde{\phi}_\pi \pi_t + \tilde{\phi}_y y_t$$

  not just ex post on eq'm path but also *in expectation*

Limit: with **many monetary shocks** we seem to recover the **true counterfactual**

Note: counterfactual rule is enforced *ex-post* and *ex-ante* using only date-0 shocks. No ex-post surprises.

Discussion and next steps

- **Expectation effects** were an obvious wedge between **truth** & **Sims-Zha estimand**
- Preceding figures: using more **policy shocks** to align expectations seemed to work
  - Specifically, we compared two experiments:
    1. Propagation of $\boldsymbol{\varepsilon}$ if new, counterfactual rule is followed
    2. Propagation of $\boldsymbol{\varepsilon}$ if old rule is followed, but it is subject to a vector of date-0 policy shocks so that new rule holds in expectation
  - They gave the same answer, so Lucas critique seemed to bite *only* via expectations.
    And so Sims-Zha $\neq$ true systematic policy counterfactual only because of ex-post surprises.
- Obvious **Q**: how general is that? what's the underlying economics?

Outline

1. History of Thought
   - Shocks vs. Rules in the Literature
   - Towards a Bridge: Sims-Zha

2. Shocks-Rules Equivalence
   - Identification Results
   - Connection to Lucas & Sims

3. Policy Causal Effects in Theory & Data

Towards a formal identification result

- **Model** [perfect foresight = 1st-order perturbation w/ aggregate risk]

$$\mathcal{H}_x \boldsymbol{x} + \mathcal{H}_z \boldsymbol{z} + \mathcal{H}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0} \quad (2)$$

$$\mathcal{A}_x \boldsymbol{x} + \mathcal{A}_z \boldsymbol{z} + \boldsymbol{\nu} = \mathbf{0} \quad (3)$$

Boldface denotes time paths, e.g. $\boldsymbol{x} = (x_0, x_1, x_2, \dots)'$. Solution of (2) - (3) = impulse responses.

Towards a formal identification result

- **Model** [perfect foresight = 1st-order perturbation w/ aggregate risk]

$$\mathcal{H}_x \boldsymbol{x} + \mathcal{H}_z \boldsymbol{z} + \mathcal{H}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0} \quad (2)$$

$$\mathcal{A}_x \boldsymbol{x} + \mathcal{A}_z \boldsymbol{z} + \boldsymbol{\nu} = \mathbf{0} \quad (3)$$

Boldface denotes time paths, e.g. $\boldsymbol{x} = (x_0, x_1, x_2, \dots)'$. Solution of (2) - (3) = impulse responses.

  - Illustrative example: three-equation NK model

Towards a formal identification result

- **Model** [perfect foresight = 1st-order perturbation w/ aggregate risk]

$$\mathcal{H}_x \boldsymbol{x} + \mathcal{H}_z \boldsymbol{z} + \mathcal{H}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0} \quad (2)$$

$$\mathcal{A}_x \boldsymbol{x} + \mathcal{A}_z \boldsymbol{z} + \boldsymbol{\nu} = \mathbf{0} \quad (3)$$

Boldface denotes time paths, e.g. $\boldsymbol{x} = (x_0, x_1, x_2, \dots)'$. Solution of (2) - (3) = impulse responses.

  - Illustrative example: three-equation NK model

$$\begin{pmatrix} 1 & -1 & 0 & \dots \\ 0 & 1 & -1 & \dots \\ 0 & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \boldsymbol{y} + \frac{1}{\gamma} \boldsymbol{i} - \frac{1}{\gamma} \begin{pmatrix} 0 & 1 & 0 & \dots \\ 0 & 0 & 1 & \dots \\ 0 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \boldsymbol{\pi} = \mathbf{0}$$

Towards a formal identification result

- **Model** [perfect foresight = 1st-order perturbation w/ aggregate risk]

$$\mathcal{H}_x \boldsymbol{x} + \mathcal{H}_z \boldsymbol{z} + \mathcal{H}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0} \quad (2)$$

$$\mathcal{A}_x \boldsymbol{x} + \mathcal{A}_z \boldsymbol{z} + \boldsymbol{\nu} = \mathbf{0} \quad (3)$$

Boldface denotes time paths, e.g. $\boldsymbol{x} = (x_0, x_1, x_2, \dots)'$. Solution of (2) - (3) = impulse responses.

  - Illustrative example: three-equation NK model

$$\begin{pmatrix} 1 & -\beta & 0 & \dots \\ 0 & 1 & -\beta & \dots \\ 0 & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \boldsymbol{\pi} - \kappa \boldsymbol{y} - \begin{pmatrix} 1 \\ \rho_\varepsilon \\ \rho_\varepsilon^2 \\ \vdots \end{pmatrix} \varepsilon_0 = \mathbf{0}$$

Towards a formal identification result

- **Model** [perfect foresight = 1st-order perturbation w/ aggregate risk]

$$\mathcal{H}_x \boldsymbol{x} + \mathcal{H}_z \boldsymbol{z} + \mathcal{H}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0} \quad (2)$$

$$\mathcal{A}_x \boldsymbol{x} + \mathcal{A}_z \boldsymbol{z} + \boldsymbol{\nu} = \mathbf{0} \quad (3)$$

Boldface denotes time paths, e.g. $\boldsymbol{x} = (x_0, x_1, x_2, \dots)'$. Solution of (2) - (3) = impulse responses.

  - Illustrative example: three-equation NK model

$$-\boldsymbol{i} + \boldsymbol{\phi}\boldsymbol{\pi} + \begin{pmatrix} \nu_0 & \nu_1 & \nu_2 & \dots \end{pmatrix}' = \mathbf{0}$$

Towards a formal identification result

- **Model** [perfect foresight = 1st-order perturbation w/ aggregate risk]

$$\mathcal{H}_x \boldsymbol{x} + \mathcal{H}_z \boldsymbol{z} + \mathcal{H}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0} \quad (2)$$

$$\mathcal{A}_x \boldsymbol{x} + \mathcal{A}_z \boldsymbol{z} + \boldsymbol{\nu} = \mathbf{0} \quad (3)$$

Boldface denotes time paths, e.g. $\boldsymbol{x} = (x_0, x_1, x_2, \dots)'$. Solution of (2) - (3) = impulse responses.

  - Illustrative example: three-equation NK model
  - More generally: many linearized models fit (2) - (3) [RBC, NK-DSGE, HANK, ...] Details

Towards a formal identification result

- **Model** [perfect foresight = 1st-order perturbation w/ aggregate risk]

$$\mathcal{H}_x \boldsymbol{x} + \mathcal{H}_z \boldsymbol{z} + \mathcal{H}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0} \quad (2)$$

$$\mathcal{A}_x \boldsymbol{x} + \mathcal{A}_z \boldsymbol{z} + \boldsymbol{\nu} = \mathbf{0} \quad (3)$$

Boldface denotes time paths, e.g. $\boldsymbol{x} = (x_0, x_1, x_2, \dots)'$. Solution of (2) - (3) = impulse responses.

  - Illustrative example: three-equation NK model
  - More generally: many linearized models fit (2) - (3) [RBC, NK-DSGE, HANK, ...] ▸ Details
  - This environment has **two key features**:

    (i) Instrument sufficiency: private sector responds only to policy instrument, not rule *per se*
    Changing $\mathcal{A}_\bullet$'s does not affect $\mathcal{H}_\bullet$'s, only effect is *via* $\boldsymbol{z}$. Full Lucas critique discussion later.

    (ii) Linearity in aggregates: can restrict attention to expected values [not sign, size, state, ...]

Towards a formal identification result

- **Model** [perfect foresight = 1st-order perturbation w/ aggregate risk]

$$\mathcal{H}_x \boldsymbol{x} + \mathcal{H}_z \boldsymbol{z} + \mathcal{H}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0} \quad (2)$$

$$\mathcal{A}_x \boldsymbol{x} + \mathcal{A}_z \boldsymbol{z} + \boldsymbol{\nu} = \mathbf{0} \quad (3)$$

Boldface denotes time paths, e.g. $\boldsymbol{x} = (x_0, x_1, x_2, \ldots)'$. Solution of (2) - (3) = impulse responses.

- **Equilibrium**: bounded $\{\boldsymbol{x}, \boldsymbol{z}\}$ that solves (2) - (3) given bounded $\{\boldsymbol{\varepsilon}, \boldsymbol{\nu}\}$
  - Assume that, under baseline rule, eq'm exists & is unique
  - Write IRFs to path $\boldsymbol{\varepsilon}$ under baseline rule – i.e., $\boldsymbol{\nu} = \mathbf{0}$ – as $\{\boldsymbol{x}(\boldsymbol{\varepsilon}), \boldsymbol{z}(\boldsymbol{\varepsilon})\}$

Towards a formal identification result

- **Model** [perfect foresight = 1st-order perturbation w/ aggregate risk]

$$\mathcal{H}_x \boldsymbol{x} + \mathcal{H}_z \boldsymbol{z} + \mathcal{H}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0} \quad (2)$$

$$\mathcal{A}_x \boldsymbol{x} + \mathcal{A}_z \boldsymbol{z} + \boldsymbol{\nu} = \mathbf{0} \quad (3)$$

  Boldface denotes time paths, e.g. $\boldsymbol{x} = (x_0, x_1, x_2, \dots)'$. Solution of (2) - (3) = impulse responses.

- **Equilibrium**: bounded $\{\boldsymbol{x}, \boldsymbol{z}\}$ that solves (2) - (3) given bounded $\{\boldsymbol{\varepsilon}, \boldsymbol{\nu}\}$
  - Assume that, under baseline rule, eq'm exists & is unique
  - Write IRFs to path $\boldsymbol{\varepsilon}$ under baseline rule – i.e., $\boldsymbol{\nu} = \mathbf{0}$ – as $\{\boldsymbol{x}(\boldsymbol{\varepsilon}), \boldsymbol{z}(\boldsymbol{\varepsilon})\}$
- **Object of interest**: IRFs $\{\tilde{\boldsymbol{x}}(\boldsymbol{\varepsilon}), \tilde{\boldsymbol{z}}(\boldsymbol{\varepsilon})\}$ if (3) is replaced by cnfct'l rule $\tilde{\mathcal{A}}_x \boldsymbol{x} + \tilde{\mathcal{A}}_z \boldsymbol{z} = \mathbf{0}$.

Dynamic causal effects

- Solving the system under the baseline rule gives

$$\begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{z} \end{pmatrix} = \underbrace{-\begin{pmatrix} \mathcal{H}_x & \mathcal{H}_z \\ \mathcal{A}_x & \mathcal{A}_z \end{pmatrix}^{-1} \times \begin{pmatrix} \mathcal{H}_\varepsilon & \mathbf{0} \\ \mathbf{0} & I \end{pmatrix}}_{\equiv \Theta} \times \begin{pmatrix} \boldsymbol{\varepsilon} \\ \boldsymbol{\nu} \end{pmatrix}, \quad \Theta \equiv \begin{pmatrix} \Theta_{\boldsymbol{x},\boldsymbol{\varepsilon}} & \Theta_{\boldsymbol{x},\boldsymbol{\nu}} \\ \Theta_{\boldsymbol{z},\boldsymbol{\varepsilon}} & \Theta_{\boldsymbol{z},\boldsymbol{\nu}} \end{pmatrix}$$

- Our graphical explorations relied on **two key inputs**:
  1. **Non-policy**: causal effects of $\boldsymbol{\varepsilon}$ under base rule $\{\boldsymbol{x}(\boldsymbol{\varepsilon}) = \Theta_{\boldsymbol{x},\boldsymbol{\varepsilon}} \times \boldsymbol{\varepsilon}, \boldsymbol{z}(\boldsymbol{\varepsilon}) = \Theta_{\boldsymbol{z},\boldsymbol{\varepsilon}} \times \boldsymbol{\varepsilon}\}$
  2. **Policy**: causal effects of all current *and news* shocks $\boldsymbol{\nu}$ to the base rule, $\{\Theta_{\boldsymbol{x},\boldsymbol{\nu}}, \Theta_{\boldsymbol{z},\boldsymbol{\nu}}\}$
     - This is multi-dimensional—each column gives the IRF to a particular policy shock
     - First column = contemp. shock, later columns = news shocks

From policy shocks to counterfactuals

Proposition

For any $\{\tilde{\mathcal{A}}_x, \tilde{\mathcal{A}}_z\}$ *that induces a unique eq'm, we can recover the counterfactuals* $\tilde{\boldsymbol{x}}(\boldsymbol{\varepsilon})$ *and* $\tilde{\boldsymbol{z}}(\boldsymbol{\varepsilon})$ *as the impulse responses under the baseline rule to the shocks* $\{\boldsymbol{\varepsilon}, \tilde{\boldsymbol{\nu}}\}$*, where* $\tilde{\boldsymbol{\nu}}$ *solves*

$$\tilde{\mathcal{A}}_x \left(\boldsymbol{x}(\boldsymbol{\varepsilon}) + \Theta_{x,\nu} \times \tilde{\boldsymbol{\nu}}\right) + \tilde{\mathcal{A}}_z \left(\boldsymbol{z}(\boldsymbol{\varepsilon}) + \Theta_{z,\nu} \times \tilde{\boldsymbol{\nu}}\right) = \mathbf{0} \tag{4}$$

In words: compute IRFs under the baseline rule $\boldsymbol{\varepsilon}$ *and an artificial date-0 policy shock vector* $\tilde{\boldsymbol{\nu}}$*, chosen so that cnfct'l rule holds following* $\{\boldsymbol{\varepsilon}, \tilde{\boldsymbol{\nu}}\}$*. Note how this is the date-0, vector-of-shocks generalization of Sims-Zha.*

- **Intuition**: private sector only cares about (expected) instrument path
  - Select policy shocks to enforce cnfct'l rule. Then same instrument $\rightarrow$ same outcomes.
  - Aside: if only first entry of $\tilde{\boldsymbol{\nu}}$ is non-zero, then this is Sims-Zha!

For the complete proof see McKay-Wolf (2023). Will turn to eq'm uniqueness next.

Equilibrium multiplicity and policy regimes

- So far assumed eq'm uniqueness under counterfactual rule. What if there's **multiplicity**?

Proposition

For any $\{\tilde{\mathcal{A}}_x, \tilde{\mathcal{A}}_z\}$ *that induces multiple equilibria, we can recover the set of all possible counterfactuals* $\tilde{\boldsymbol{x}}(\boldsymbol{\varepsilon})$ *and* $\tilde{\boldsymbol{z}}(\boldsymbol{\varepsilon})$ *by computing the set of all solutions* $\tilde{\boldsymbol{\nu}}$ *of*

$$\tilde{\mathcal{A}}_x \left(\boldsymbol{x}(\boldsymbol{\varepsilon}) + \Theta_{x,\nu} \times \tilde{\boldsymbol{\nu}}\right) + \tilde{\mathcal{A}}_z \left(\boldsymbol{z}(\boldsymbol{\varepsilon}) + \Theta_{z,\nu} \times \tilde{\boldsymbol{\nu}}\right) = \mathbf{0}, \tag{5}$$

and then evaluating the impulse responses under the baseline rule to $\{\boldsymbol{\varepsilon}, \tilde{\boldsymbol{\nu}}\}$.

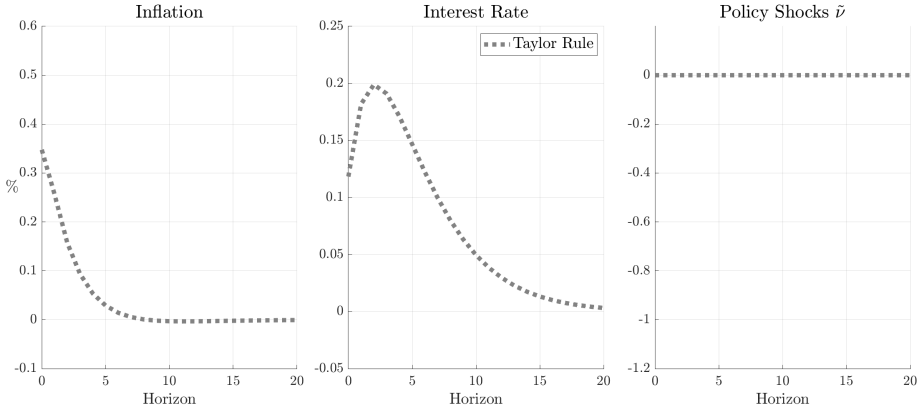
- Economic intuition
  - All cnfct'l equilibria: private-sector block still holds, just now have *many* bounded solutions
  - Knowledge of Θ_ν is enough to trace out all of them via shocks $\tilde{\nu}$, then just need to **select**
- Opens a second role for policy: select among the many possible equilibria

  E.g., FTPL: select unique eq'm in which gov't budget constraint holds (see Cochrane, 2011 & 2023).

Monetary vs. fiscal dominance

Illustration: monetary vs. fiscal dominance in a textbook RANK model

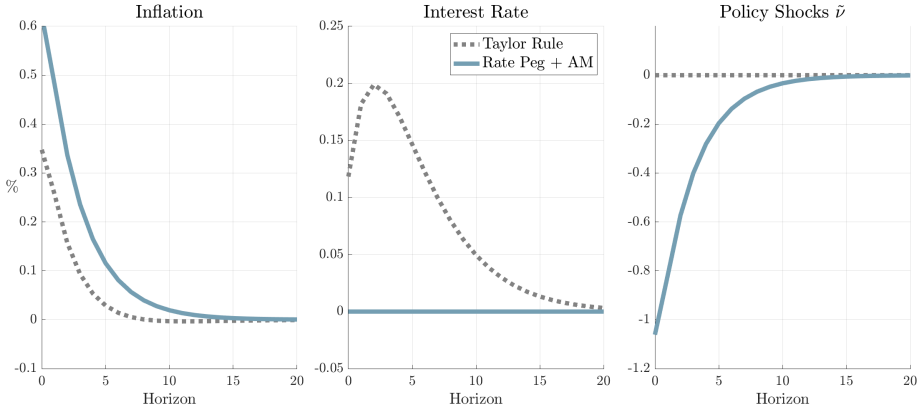
Study cnfct'l propagation of cost-push shock under rate peg, using monetary vs. fiscal selection.



Monetary vs. fiscal dominance

Illustration: monetary vs. fiscal dominance in a textbook RANK model

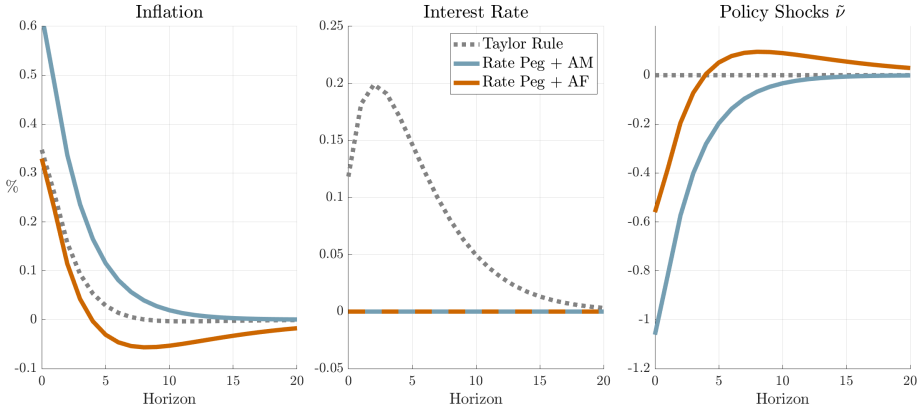
Study cnfct'l propagation of cost-push shock under rate peg, using monetary vs. fiscal selection.



Monetary vs. fiscal dominance

Illustration: monetary vs. fiscal dominance in a textbook RANK model

Study cnfct'l propagation of cost-push shock under rate peg, using monetary vs. fiscal selection.



Extension to other counterfactuals

- Can we extend this logic to our other two counterfactuals?
  - Suppose that economy is subject to full vector of (policy + non-policy) shocks e_t, giving

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell e_{t-\ell}, \quad \text{Var}(e_t) = I \tag{6}$$

    where the y_t's are observable macroeconomic outcomes, $y_t = (x_t', z_t')'$
    For each shock, the IRFs Θ_ℓ come from a linear perfect-foresight system like (2) - (3).
  - Contemplate changing the policy rule to $\tilde{A}_x \boldsymbol{x} + \tilde{A}_z \boldsymbol{z} = \boldsymbol{0}$.

Extension to other counterfactuals

- Can we extend this logic to our other two counterfactuals?
  - Suppose that economy is subject to full vector of (policy + non-policy) shocks e_t, giving

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell e_{t-\ell}, \quad \text{Var}(e_t) = I \tag{6}$$

    where the y_t's are observable macroeconomic outcomes, $y_t = (x_t', z_t')'$
    For each shock, the IRFs Θ_ℓ come from a linear perfect-foresight system like (2) - (3).
  - Contemplate changing the policy rule to $\tilde{A}_x \boldsymbol{x} + \tilde{A}_z \boldsymbol{z} = \boldsymbol{0}$. Then:

    Q2 **Business-cycle statistics**: cnfct'l IRFs $\tilde{\Theta}_\ell$ give cnfct'l autocovariance function

$$\tilde{\Gamma}_y(\ell) = \sum_{m=0}^{\infty} \tilde{\Theta}_{m+\ell} \tilde{\Theta}_m'$$

Extension to other counterfactuals

- Can we extend this logic to our other two counterfactuals?
  - Suppose that economy is subject to full vector of (policy + non-policy) shocks e_t, giving

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell e_{t-\ell}, \quad \text{Var}(e_t) = I \tag{6}$$

    where the y_t's are observable macroeconomic outcomes, $y_t = (x_t', z_t')'$
    For each shock, the IRFs Θ_ℓ come from a linear perfect-foresight system like (2) - (3).
  - Contemplate changing the policy rule to $\tilde{A}_x \boldsymbol{x} + \tilde{A}_z \boldsymbol{z} = \boldsymbol{0}$. Then:

    Q3 **Alternative history**: cnfct'l outcomes at a particular date t would be [ignoring initial cond's]

$$\tilde{y}_t = \sum_{\ell=0}^{\infty} \tilde{\Theta}_\ell e_{t-\ell}$$

Extension to other counterfactuals

- Can we extend this logic to our other two counterfactuals?
  - Suppose that economy is subject to full vector of (policy + non-policy) shocks e_t, giving

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell e_{t-\ell}, \quad \text{Var}(e_t) = I \tag{6}$$

    where the y_t's are observable macroeconomic outcomes, $y_t = (x_t', z_t')'$
    For each shock, the IRFs Θ_ℓ come from a linear perfect-foresight system like (2) - (3).
  - Contemplate changing the policy rule to $\tilde{A}_x \boldsymbol{x} + \tilde{A}_z \boldsymbol{z} = \boldsymbol{0}$. Then:

    Q3 **Alternative history**: cnfct'l outcomes at a particular date t would be [ignoring initial cond's]

$$\tilde{y}_t = \sum_{\ell=0}^{\infty} \tilde{\Theta}_\ell e_{t-\ell}$$

  - Immediate if we can observe e_t, just proceed shock-by-shock. Can be relaxed; we will do so in Lecture Note 9.

Outline

1. History of Thought
   - Shocks vs. Rules in the Literature
   - Towards a Bridge: Sims-Zha

2. Shocks-Rules Equivalence
   - Identification Results
   - Connection to Lucas & Sims

3. Policy Causal Effects in Theory & Data

Discussion

- Results apply to a large **class of macro models**. Where's the **Lucas critique**?
  Recall key asn's: instrument sufficiency & linearity. Consistent with typical RANK, HANK, ...
  - Traditional formulation is in state-space:

$$\begin{aligned} s_t &= A(\bullet)s_{t-1} + B(\bullet)e_t \\ y_t &= C(\bullet)s_{t-1} + D(\bullet)e_t \end{aligned}$$

    Famous Hansen-Sargent cross-equation restrictions: $\{A, B, C, D\}$ all depend on policy rule
  - Here: policy invariance is at a more primitive level, for the $\mathcal{H}(\bullet)$'s
    Lucas critique bites via instrument paths $\boldsymbol{z}$, not by changing the $\mathcal{H}(\bullet)$ coefficients.
- There are still some meaningful limitations:
  1. Instrument sufficiency fails in some environments, e.g., in signal extraction problems
     E.g., in Lucas island economy rule matters above and beyond nominal demand growth.
  2. Linearity is for practical feasibility, allows us to later more easily connect with data.

Summary: the dual roles of policy

- Zooming out: systematic policy in the **impulse-propagation paradigm** plays two roles
  1. On-equilibrium path of policy, enters the private-sector block as $\mathcal{H}_z \boldsymbol{z}$
  2. Selection among the many possible equilibria consistent with the path $\boldsymbol{z}$
- We have seen: **policy shock** causal effects are enough to switch to any other rule
  - Delivers the stated revision of the impulse-propagation paradigm: inputs ("impulses") are now non-policy shocks + policy instrument path + policy eq'm selection
  - Vindication of Sims-Zha: shocks are very useful, just need higher-dimensional space
- Next: prepare transition to **empirics**

Outline

1. History of Thought
   - Shocks vs. Rules in the Literature
   - Towards a Bridge: Sims-Zha

2. Shocks-Rules Equivalence
   - Identification Results
   - Connection to Lucas & Sims

3. Policy Causal Effects in Theory & Data

Policy causal effects in theory & data

- To connect with empirics, let's begin by studying the role of the initial rule $\mathcal{A}_x, \mathcal{A}_z$
  - Θ_ν is defined relative to that rule. But argument would also work if we observed shock effects relative to a different rule. Rule is the "basis" w.r.t. which causal effects are defined.
  - What if the rule was $\mathcal{A}_x = \mathbf{0}$, $\mathcal{A}_z = -I +$ some eq'm selection? Then, heuristically:

$$x = \underbrace{-\mathcal{H}_x^{-1}\mathcal{H}_z}_{\equiv \Theta_{x,z}} \times z$$

  $\Theta_{x,z}$ gives the causal effects of any instrument *path* on macro outcomes. The basis is now not an arbitrary rule, but the policy instrument itself. **Shock space** vs. **instrument space**.

- Next two slides:
  1. Quick illustration of the **basis change** using the familiar 3-equation NK model
  2. Preview of **empirics**

NK model illustration

- Consider a simple NK model with iid shocks:

$$y_t = -\frac{1}{\gamma}(i_t - \underbrace{\mathbb{E}_t\pi_{t+1}}_{=0}) + \underbrace{\mathbb{E}_t y_{t+1}}_{=0}, \quad \pi_t = \kappa y_t + \beta \underbrace{\mathbb{E}_t\pi_{t+1}}_{=0}, \quad i_t = \phi_\pi \pi_t + \nu_t$$

Note: exact return to steady state embeds monetary dominance (Angeletos-Lian-Wolf-Zhang, 2026).

- Solving for the mapping to endogenous outcomes:
  - In shock space:

$$y_t = -\frac{1}{\gamma + \phi_\pi \kappa}\nu_t, \quad \pi_t = -\frac{\kappa}{\gamma + \phi_\pi \kappa}\nu_t, \quad i_t = \frac{\gamma}{\gamma + \phi_\pi \kappa}\nu_t$$

  - In instrument space:

$$y_t = -\frac{1}{\gamma} i_t, \quad \pi_t = -\frac{\kappa}{\gamma} i_t$$

Note how the instrument space mapping depends only on the private-sector block, not the policy rule.

Towards empirical strategies

- Recall our general monetary policy rule from before:

$$i_t = f(\Omega_t) + \nu_{0,t} + \sum_{\ell=1}^{\infty} \nu_{\ell,t-\ell}$$

- Today's theoretical discussion suggests a **two-step empirical strategy**:

  1. Find a way to measure the ν's, or some linear combination of them, or an IV for them
     Intuition: movement in policy that is orthogonal to all the other stochastic drivers of the economy. A true exogenous policy impulse.
  2. Regress both z and x on those ν's to identify (parts of) $\Theta_{x,z}$

     → z regression gives the "treatment", x regression gives the treatment effects [= applied micro]

     → Note: IRFs will embed both instrument causal effects + in-sample eq'm selection
     Why? shock(s) to in-sample rule induce this z path and embed that rule's selection.

- We will start on this path in the next set of lecture notes ...

Summary & takeaways

This lecture: policy & policy causality in the impulse-propagation paradigm

- Key idea: equivalence of **policy rules & shocks**
  - Policy acts through two levers: path of instrument + eq'm selection
  - Thus there's a well-defined notion of **policy causal effects**, $\Theta_{x,z}$
- Looking ahead: our job is now to learn about $\Theta_{x,z}$
  1. Lecture Notes 5-6: time-series strategies
  2. Lecture Note 9: using model structure to supplement the empirics
  3. Lecture Notes 11-12: cross-sectional strategies

Model examples

1. **HANK model**
   - Generalized IS curve

$$\boldsymbol{c} = \mathcal{C}(\boldsymbol{y}, \boldsymbol{\pi}, \boldsymbol{i}, \boldsymbol{\varepsilon}^d) = \mathcal{C}_y \boldsymbol{y} + \mathcal{C}_\pi \boldsymbol{\pi} + \mathcal{C}_i \boldsymbol{i} + \boldsymbol{\varepsilon}^d$$

2. **Behavioral models**
   - Various behavioral frictions correspond to simple adjustments of the matrices in $\mathcal{H}$
   - Example: sticky information in consumption decisions, for any decision input p

$$\tilde{\mathcal{C}}_p(t,s) = \sum_{q=1}^{\min(t,s)} [\mathcal{E}(q,s) - \mathcal{E}(q-1,s)] \mathcal{C}_p(t-q+1, s-q+1)$$

   where $\mathcal{E}(0,s) = 0$ and

$$\mathcal{E} = \begin{pmatrix} 1 & 1-\theta & 1-\theta & \dots \\ 1 & 1 & 1-\theta^2 & \dots \\ 1 & 1 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

back